THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 5: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A chord is at distance 7 cm from the centre of a circle of radius 25 cm. Its length (in cm) is: 1

(a) 48 **(b)** 47 **(c)** 49 **(d)** 50

2. The radius of a circle is 25 cm and a chord is 40 cm long. The distance of the chord from the centre (in cm) is: 1

(a) 16 **(b)** 14 **(c)** 17 **(d)** 15

3. The chord AB subtends 74° at the centre O. The angle subtended at a point on the major arc is: 1

(a) 37° **(b)** 47° **(c)** 27° **(d)** 57°

4. The chord AB subtends 66° at the centre. The angle it subtends at a point on the minor arc is: 1

(a) 157° **(b)** 167° **(c)** 147° **(d)** 137°

5. In a cyclic quadrilateral $ABCD$, $\angle A = 55^\circ$ and the side BC is produced to E. Then $\angle DCE =$ 1

(a) 75° **(b)** 55° **(c)** 65° **(d)** 45°

6. In triangle OAB, $OA = OB$ and $\angle AOB = 80^\circ$ (O is the centre). Then $\angle OAB =$ 1

(a) 70° **(b)** 40° **(c)** 50° **(d)** 60°

7. Equal chords of a circle are ___ from the centre. 1

(a) equidistant **(b)** at distances in ratio $1 : 2$

(c) on the same line **(d)** at different distances

8. A chord is at distance 24 cm from the centre of a circle of radius 25 cm. Its length (in cm) is: 1

(a) 13 **(b)** 14 **(c)** 16 **(d)** 15

9. The radius of a circle is 25 cm and a chord is 30 cm long. The distance of the chord from the centre (in cm) is: 1

(a) 19 **(b)** 21 **(c)** 22 **(d)** 20

10. The chord AB subtends 88° at the centre O. The angle subtended at a point on the major arc is: 1

(a) 44° **(b)** 64° **(c)** 54° **(d)** 34°

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In a circle of radius 26 cm, a chord is 10 cm from the centre. Find its length. 2

12. A chord AB of a circle with centre O subtends an angle of 60° at O. Find the angle it subtends at a point C on the minor arc. 2

13. $ABCD$ is a cyclic quadrilateral and BC is produced to E. If $\angle A = 62^\circ$, find $\angle DCE$. 2

14. In a circle of radius 26 cm, a chord is 24 cm from the centre. Find its length. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In triangle OAB with O the centre of a circle, $\angle AOB = 70^\circ$. Find $\angle OAB$ and the angle subtended by AB at a point of the major arc. 3

16. In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 10)^\circ$ and $\angle C = (3x - 5)^\circ$. Find x and $\angle A$. 3

17. A chord of length 30 cm is 8 cm from the centre of a circle. Find the radius and the area of the right triangle formed by the radius, the perpendicular from the centre and half the chord. 3

OR

In triangle OAB with O the centre of a circle, $\angle AOB = 90^\circ$. Find $\angle OAB$ and the angle subtended by AB at a point of the major arc. 3

18. In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 15)^\circ$ and $\angle C = (3x - 10)^\circ$. Find x and $\angle A$. 3

OR

A chord of length 16 cm is 15 cm from the centre of a circle. Find the radius and the area of the right triangle formed by the radius, the perpendicular from the centre and half the chord. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two parallel chords of lengths 10 cm and 24 cm lie on the same side of the centre of a circle of radius 13 cm. Find the distance between them. 5

OR

$ABCD$ is a cyclic quadrilateral with $\angle A = 75^\circ$, $\angle B = 88^\circ$ and side BC is produced to E, side AB to F. Find $\angle C$, $\angle D$, $\angle DCE$ and $\angle DAF$. 5

20. Two parallel chords of lengths 16 cm and 30 cm lie on the same side of the centre of a circle of radius 17 cm. Find the distance between them. 5

OR

$ABCD$ is a cyclic quadrilateral with $\angle A = 66^\circ$, $\angle B = 94^\circ$ and side BC is produced to E, side AB to F. Find $\angle C$, $\angle D$, $\angle DCE$ and $\angle DAF$. 5

END OF QUESTION PAPER — **Total: 40 Marks**